Yiwei Ding

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EDUCATION

PhD, Computational Chemistry, **University of Chicago** Sep 2022 – Dec 2027 (expected)
Seymour Goodman Fellowship; Relevant courses: **Machine Learning**, Introduction to Probabilistic Models
BS, Physics, **University of Science and Technology of China** Sep 2018 – Jun 2022

WORK EXPERIENCE

Quant Researcher Intern| Deep Data Capital Aug 2026 – Oct 2026

- Built an end-to-end return-prediction pipeline in Python (scikit-learn, pandas, NumPy) to forecast next-day stock returns from a ~5M-row, multi-year Chinese A-share panel dataset spanning 2016–2021.
- Engineered 20+ predictive factors, including adjustment-corrected momentum and volatility signals, cross-sectional price/volume ratios, and market microstructure indicators (daily price-limit status), resolving ambiguous data conventions through empirical variance-minimization testing.
- Designed a leakage-safe, date-based expanding-window cross-validation framework to prevent lookahead bias, a common failure mode in financial machine learning.
- Tuned a gradient boosting model via grid search, achieving an out-of-sample correlation of 0.065 and IC of 0.031 on a 1.9M-row holdout — 20x+ over baseline.

Machine Learning Engineer Intern | Eth Tech Aug 2025 – Oct 2025

- Built and deployed a natural language search system on **AWS** using **Elasticsearch**, supporting keyword, semantic, and time-based queries across text, image, audio and video content.
- Designed and implemented multimedia indexing pipelines leveraging **BLIP**, **OCR**, and **Whisper** to extract semantic data from images, audios and video files for downstream retrieval.
- Developed a robust query processing workflow that translates free-form user queries into structured search tools calls, orchestrated using **LLMs** and **LangChain**.
- Applied **retrieval augmented generation (RAG)** to enable direct question answering over indexed content.
- Fine-tuned LLMs to improve query translation quality and search relevance, resulting in a **17% relative improvement in Recall@5 and a 4% improvement in MRR.**
- Optimized model inference performance through **post-training quantization**, **LLM runtimes (vLLM/TGI)**, **prefix caching** and **speculative decoding** to reduce latency and improve throughput.

LLM Research Intern | TikTok (ByteDance), China Aug 2024 – Oct 2024

- Researched continual training of large language models, focusing on updating models with new data without full retraining.
- Applied parameter-efficient fine-tuning (LoRA) to adapt pretrained LLMs while limiting forgetting of previously learned capabilities.
- Built and ran benchmark evaluations comparing updated models against baselines on both new-task performance and retention of prior performance.
- Analyzed catastrophic forgetting through training and validation loss curves, studying how it appeared during model updates.
- Investigated the effect of key hyperparameters (e.g., learning rate, LoRA rank) on the tradeoff between learning new information and retaining old knowledge.
- Implemented experiments in PyTorch and collaborated with a team of researchers.
- Wrote a report summarizing methods and findings, and proposed directions for future work.

PUBLICATIONS

A Theoretical Framework to Interpret the Escape Time of a Bound Ligand from Random Accelerated Molecular Dynamics Simulations
Yiwei Ding, *Alessia Ghidini, Pilar Cossio, Benoit Roux*
J. Chem. Phys. 164, 064109 (2026)

Biaxial stretch-induced crystallization of polymers: A molecular dynamics simulation study
Cui Nie, Fan Peng, Tingyu Xu, ***Yiwei Ding****, Junfang Sheng, Wei Chen, Liangbin Li*
Macromolecules 2021, 54, 21, 9794–9803

Deep learning model for committor-based transition pathway from non-equilibrium steered MD
Yiwei Ding*, Christophe Chipot, Benoit Roux*
In preparation

SKILLS

Programming Languages: Python, Matlab, R
AI/ML: Deep Learning, NLP, Generative AI, LLM, RAG, Ranking
Tools & Frameworks: PyTorch, TensorFlow, Elasticsearch, Git, LangChain